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THE EXTENT OF GREATER INDIA, I. PRELIMINARY PALAEOMAGNETIC DATA FROM THE UPPER DEVONIAN OF THE EASTERN HINDUKUSH, CHITRAL (PAKISTAN)

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Samples of Upper Devonian sedimentary ironstones from the eastern Hindukush, Chitral (Pakistan), give a characteristic palaeomagnetic direction: declination $D = 318°$, inclination $I = -6.5°$; believed to represent the primary magnetization direction. The samples come from an area which lies north of a major ophiolite zone that recent workers suggest is the southwestern continuation of the Indus Suture. As the present palaeomagnetic results are in fair agreement with palaeomagnetic data from the Siberian platform but not with data from Gondwanaland they can be taken as additional evidence that this suture does indeed constitute the main collision zone between the Gondwanic Indian subcontinent and Asia. The palaeomagnetic data presented here from the Devonian of Chitral suggests additionally: (1) in excess of 100° of counterclockwise rotation of the area, associated most likely with the formation of the regional Hindukush-Pamir-Karakoram syntaxial bend; (2) more than 2000 km of crustal shortening between Chitral and the Siberian platform due to the northward indentation of the Indian Gondwanaland fragment subsequent to collision.

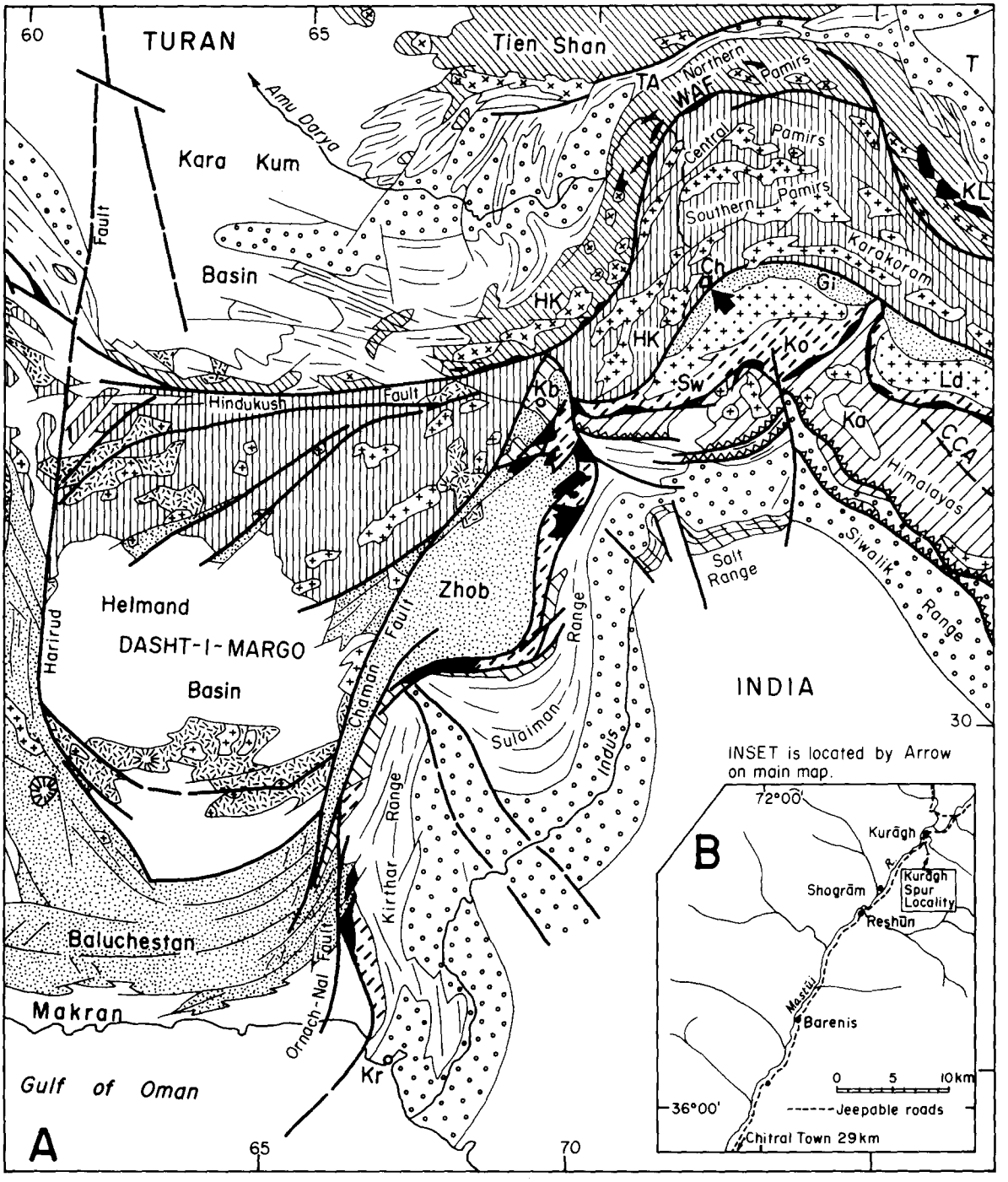
1. Introduction

Recent palaeomagnetic data have contributed importantly to the reconstruction and delineation of the original northern boundary of Gondwanaland. For example, data from central Iran [1,2], and from Afghanistan west of the Chaman fault [3] have shown that large areas of these countries north of the Zagros-Makran lineament comprise fragments of Gondwanaland. Originally, these fragments probably occupied positions at the outer rim of Gondwanaland [4,5]. In contrast, palaeomagnetic data from the Malay peninsula [6–8] and from West Kalimantan [9] show no evidence that these parts of the Sundaland extension of southeast Asia have Gondwana affinities.

The Gondwanaland affinity of India is of course now firmly established on palaeomagnetic grounds (see review articles by Klootwijk [5,10]). However, the extent of the Indian Gondwana fragment north of the Himalayas is unconstrained palaeomagnetically and has therefore been the subject of considerable speculation and debate (see review by Powell and Conaghan [11]).

2. Configuration and extent of Gondwanic continental fragments in southern and central Asia and concepts of a Greater India

Recent models of the pattern of fusion of Gondwanic elements east of Afghanistan in the



ABBREVIATIONS

Ch	Chitral	Ka	Kashmir	Ld	Ladakh
CCA	Central Crystalline Axis of Himalayas	Kb	Kabul	Sw	Swat
Gi	Gilgit	Ko	Kohistan	T	Tarim Basin
HK	Hindukush	KL	Kun - Lun	TA	Transalai
		Kr	Karachi	WAF	Wanch - Akbaytal Fault

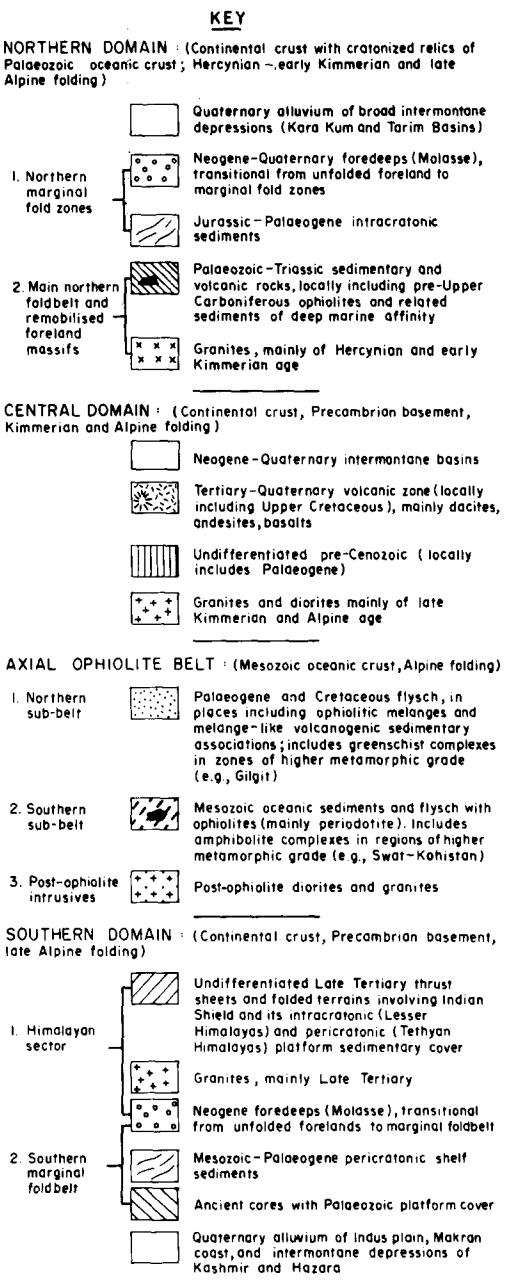


Fig. 1. Regional geological and tectonic context of the study area. A. Classification of the regional tectonic zones according to Stöcklin [12, fig. 1]. Location of study area (arrowed) is towards top-right corner of map. B. Location map of the sampled area.

Indian sector of southeastern and central Asia can be divided into two categories:

(1) Models that recognize the Axial Ophiolite Belt of Stöcklin [12] (see Fig. 1) to define the farthest exposed extent of the Indian continental fragment within Asia and that find little or no compelling evidence on which to advocate the presence of other exposed Gondwanic blocks north of this line.

(2) Models that advocate exposed Gondwanic elements to extend well north of the Axial Ophiolite Belt, either as coherent extensions of a much larger sized Indian subcontinent, or as a separate zone of amalgamated small continental blocks which coalesced with southern Angaraland by early Jurassic times prior to the arrival there in the late Cretaceous or Palaeogene of an Indian block defined as in (1).

2.1. Models of category 1

The majority of workers [11–23] identify the semi-continuous ophiolite belt (Fig. 1) which girdles the Indian subcontinent on the west (Baluchistan-Swat), north (Indus-Tsangpo lineament), and east (Arakan Yoma line) as delineating the distant edge of the Indian continental fragment within Asia and broadly accept a closing of the suture defined by this ophiolite zone in the period from latest Cretaceous to Palaeogene. There is also wide agreement that the visible or exposed areal extent of this Indian continental fragment has been reduced through crustal shortening across the Himalayan central crystalline axis (Fig. 1) following the development there in the Oligocene of a deep crustal fracture and the subsequent underthrusting of peninsular India beneath the Himalayas and possibly also beneath Tibet.

The actual extent to which India has underthrust the Himalayas and, in particular, Tibet, is controversial [11,23,24], and is not tightly constrained by presently available evidence from the Himalayas themselves though Gansser [16] estimates a minimum of 300–400 km of crustal shortening there on the basis of stratigraphic and structural considerations. Still larger estimates of the possible amount of underthrusting arise from considerations of the likely former size and shape of the Indian continental fragment *within Gondwanaland*. Thus, recognition that India's present visible eastern and northeastern extent

(i.e., as defined by the Axial Ophiolite Belt) is too small in the Smith and Hallam [25] Gondwanaland reassembly (even as modified by Griffiths' [26] alteration to the Australian-Antarctic fit) stems independently from seafloor evidence off Australia's western margin [20,27–30] as well as from facies analysis of the Palaeozoic and Mesozoic stratigraphic record of both the onshore and offshore parts of the rifted continental margin [20,31,32].

The enlarged Indian continental fragment that is predicted by, and whose northeastern extent and configuration can be reconstructed from this evidence has been called "Greater India" [20,28]. Translated north without distortion to conform with peninsular India's present position relative to Asia, Greater India's reconstructed northeastern margin is found to correspond closely in location, shape, and extent with the eastern margin of the Tibetan Plateau [20]. This co-incidence, together with: (1) palaeobiogeographic evidence that argues against Tibet itself being of Gondwanic affinity [33–35]; (2) geophysical evidence [36–38] which suggests the Himalayas and Tibet to be underlain by twice the normal 35 km of crustal thickness; and (3) geological evidence from the Himalayas that is believed to be consistent with very large-scale Neogene crustal shortening across the axial Himalayan Main Central Thrust [11,18,39], has prompted the suggestion that India has underthrust the Himalaya and central Asia to the distant edge of the Tibetan plateau [11,18–20]. This scheme leads to estimates of underthrusting of Greater India beneath central Asia in the order of 700 km and 1500 km in areas north of the central-western Himalaya (eastern Punjab Himalaya) and the eastern Himalaya respectively [11,18,19].

2.2. Models of category 2

Though not substantiated by the palaeobiogeographic evidence [33–35], several workers have speculated that other parts of central Asia north of the Indus-Tsangpo suture comprise former parts of Gondwanaland.

For example, Crawford [40,41] suggests the Indus-Tsangpo suture to have been a mantle-deep fracture of wholly intracontinental character and which formed a narrow eastward-terminating V-shaped embayment within an extended India that also comprised a contiguous Hindukush-Pamir-Tibetan lobe as well as incorporating a loosely attached Tarim block to the north. In this scheme, as well as that of Kaila and Narain [42], the northern boundary of this semi-composite enlarged Indian subcontinent lies within the Tien Shan.

A second and more recent view is that of Stöcklin [12] and Andrieux and Brunel [43] who consider that the broad arcuate belt between the Axial Ophiolite Belt in the south and the Hindukush-Wanch-Akbaytal fault lineament in the north (Fig. 1) constitutes an amalgamation of small continental blocks of Gondwanic origin. This suggestion stems from their respective tectonic analyses and classification of the wider mountain region between Iran and the western Himalaya, and from both palaeomagnetic [1–3] and palaeobiogeographic evidence [35] that in Iran and west of Kabul in Afghanistan such an origin can indeed be demonstrated for areas which lie between these two major fracture systems (i.e., between the Axial Ophiolite Belt and the Hindukush fault; see Fig. 1). Evidence, limited to the Iranian end of this belt, leads Stöcklin [4,12] to suggest that the microcontinents in this zone had already coalesced with Eurasia by the Early Jurassic following their separation from Gondwanaland during the late Palaeozoic or early Mesozoic. Stöcklin's [12] and Andrieux and Brunel's [43] suggested eastward extension of this zone of Gondwanic microcontinents across the Chaman fault at Kabul (Fig. 1) into the eastern Hindukush and Southern and Central Pamir and through the Karakoram into Tibet is entirely speculative and, though not consistent with the palaeobiogeographic evidence [33–35], could be argued to find possible support from two other lines of recent evidence bearing on the geology of Tibet. Firstly, Gansser's [17] regional analysis of ERTS photography of central Asia demonstrates Tibet north of the Indus-Tsangpo suture to be a composite mosaic of first-order structural blocks separated by major lineaments which have trends both concordant and discordant to the Indus suture. The lineaments include some which are evidently ophiolitic branches or splays of the Indus-Tsangpo suture and which in places evidently transversely disrupt as well as latitudinally separate the southern (i.e., Transhimalayan) highlands from the more northern parts of Tibet. Secondly, a pattern of radiometric age dates from Tibet

which are said to span the Phanerozoic but to young progressively southwards, and to reflect the temporal migration of orogenic activity in that direction, led Chang and Cheng [44] to postulate the episodic southward growth of Tibet due to the successive collision of small continental blocks during the Palaeozoic and Mesozoic prior to the arrival of India and the associated closure of the Indus suture. The latter scheme is somewhat similar to that of Stöcklin [12] and Andrieux and Brunel [43] although Chang and Cheng [44] are not explicit on the possible Gondwanic origin of their postulated microcontinents. Unfortunately, the radiometric data on which this scheme is based are insufficiently documented to allow appraisal of the model.

2.3. Palaeomagnetic tests of the models

Palaeomagnetic resolution of the presence of Gondwanic elements north of the Indus-Tsangpo suture requires data from rocks of the latter region that are preferably older than the postulated episodes of fragmentation and dispersal of such continental blocks from Gondwanaland. For the Gondwanic microcontinents in Iran and Afghanistan and for Gondwanic India, these times are postulated to be respectively late Palaeozoic [4,5,12,23] and Early Cretaceous [20,27–30].

Unfortunately, the sole palaeomagnetic data yet published from the region immediately north of the Indus-Tsangpo suture [45] come from rocks that are stratigraphically too young to be useful for the above test. These data – from an area in south-central Tibet east of Lhasa – give a palaeolatitude of 8°N for a period of unresolved duration in the interval late Cretaceous–early Tertiary. Since the latter interval was characterized by a very rapid northward motion of India into this approximate latitude, the Tibetan data are, moreover, unable to confidently discriminate between the Eurasian versus Indian affinity of the sampled locality.

The purpose of the present paper is to present preliminary palaeomagnetic data from an area located north of the Indus suture which we believe are capable of discriminating the Gondwanic versus Eurasian affinity of this critical region. The results reported here from an area of Upper Devonian rocks in Chitral, northwest Pakistan (Fig. 1), stem from a small sample collection made in September 1975 by Conaghan in the course of regional structural and stratigraphic studies in the eastern Hindukush. A more extensive sampling programme designed to extend these initial results has subsequently been made by Klootwijk in the same area during August 1977.

3. Geological setting and sampling

The palaeomagnetic results reported here come from a belt of Devonian sedimentary rocks that extend northwards along the valley of the Mastūj River in vicinity of the villages of Reshūn and Kurāgh (72.5°E, 36.2°N), Chitral District, Pakistan (Fig. 1). In terms of Stöcklin's [12] recent tectonic classification of the wider region, the study area lies within but towards the southern edge of his "Central Domain": that is, within the zone which separates the "Axial Ophiolite Belt" in the southeast and the "Northern Domain" of remobilised Hercynian massifs in the northwest and north, the latter domain comprising the Northern Pamir and the Tien Shan (Fig. 1).

The area is located on the western arm of the northward-convex syntaxial bend which affects the wider western Himalayan region. Regional structural trends in the study area are consequently aligned northeast-southwest, as reflected also by the pattern of major drainage and relief. The Chitral sector of Stöcklin's "Central Domain" comprises predominantly pre-Cretaceous Phanerozoic sedimentary rocks of marine platform-facies character together with, at deeper structural levels, extensively developed sedimentary terrains of black slate and associated quartz-flysch, the latter association being allegedly of Permian or Permo-Carboniferous age [46,47]. The region is extensively dislocated by a major system of mainly northwesterly- and northerly-dipping thrust faults that parallel the northward-convex regional tectonic grain, and which bring about a pattern of north-over-south or northwest-over-southeast relative tectonic movement. In some thrust slices, the circumstance of tectonically-juxtaposed sedimentary successions of similar or overlapping geological ages but of markedly contrasted relative stratigraphic thickness and gross facies character is suggestive of lateral tectonic transportation on a very large scale [48,49].

The particular thrust slice of Devonian rocks at Kurāgh from which the present samples were ob-

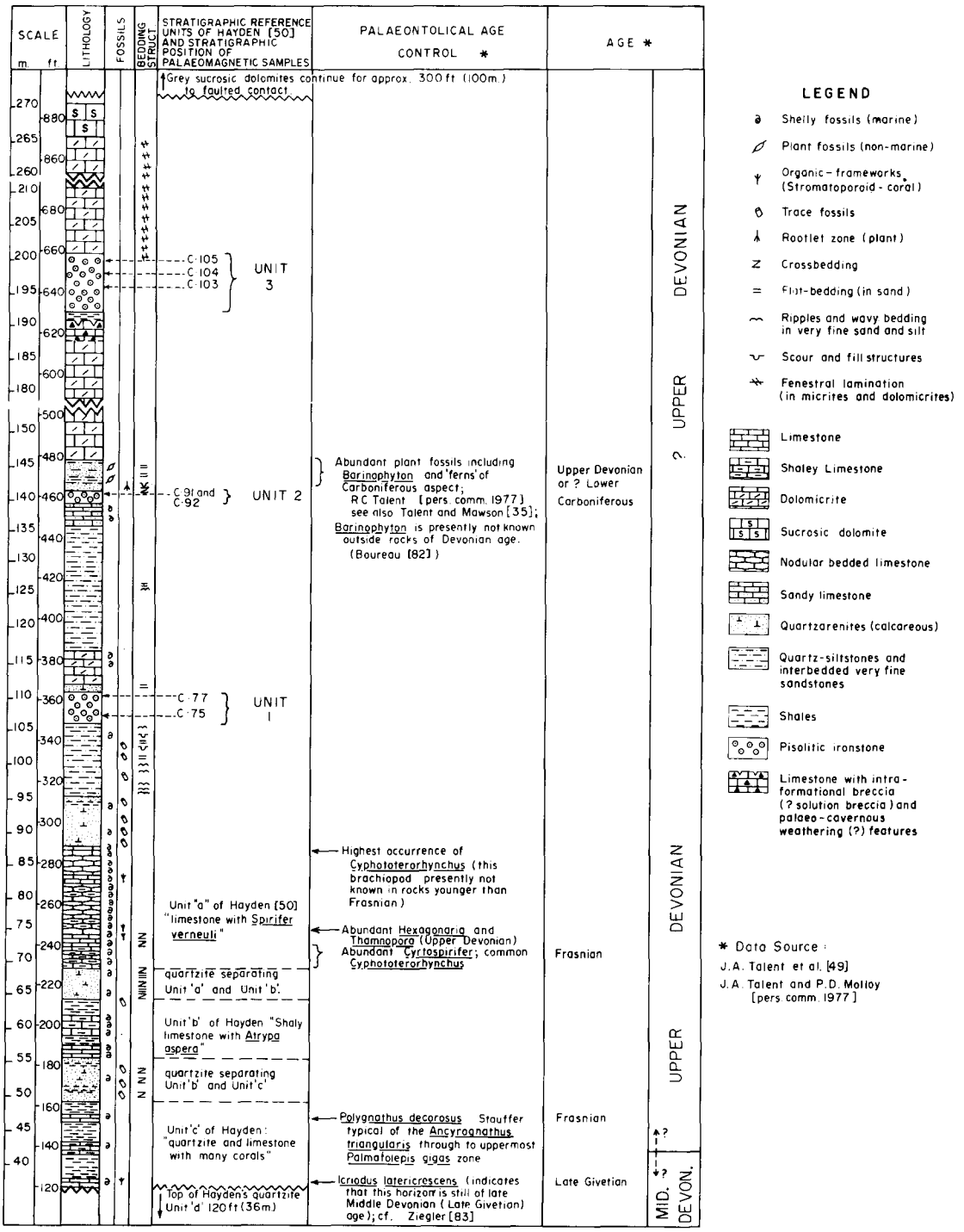


Fig. 2. Stratigraphic log of the Upper Devonian part of the succession exposed in the Kurāgh Spur, Chitral, showing the position and stratigraphic context of the three pisolitic sedimentary ironstone units analysed palaeomagnetically in this paper (see Table 1). Thicknesses are referred to the top of Hayden's (quartzite) unit "d" (see Hayden [50, plate 27, fig. 2]) and are approximate. Section logged by Conaghan, August-September 1975.

tained is structurally composite (see [49, Figs. 5 and 7]), but is dominated by a stratigraphically and structurally intact 900-m succession of Middle and Upper Devonian sedimentary rocks of predominantly shallow marine aspect (Fig. 2). This thrust sheet is juxtaposed to the southeast against another comprising a succession of sedimentary rocks of Early Devonian through Permo-Triassic age, and to the northwest against Lower Devonian slates (see [49, fig. 7]). Throughout the area the structural attitude of bedding within the various thrust sheets parallels that of the major thrust faults (i.e., dips are predominantly towards the northwest) and stratigraphic facing is in general normal. The Devonian succession in the thrust sheet at Kurāgh consists largely of competent rocks, predominantly carbonates and quartzites which, though steeply tilted and locally dislocated by minor faults and folds, are not otherwise penetratively deformed or detectably metamorphosed. The unmetamorphosed nature of the section is witnessed by the absence of greenschist grade metamorphic minerals and by the excellent preservation of shelly fossils (with unrecrystallized skeletal fabrics) and plant fossils at different stratigraphic levels (Fig. 2).

The sampled interval occurs in the upper part of the succession in a fully exposed ridge section, the "Kurāgh Spur", on the left bank of the Mastūj River overlooking Kurāgh (Fig. 1B), a classical section long renowned for its abundance of well-preserved Upper Devonian fossils [50]. The samples were collected from three sedimentary chamositic ironstone units that occur at separate levels within an interval of approximately 100 m beginning about 60 m above the base of the Upper Devonian part of the succession (Fig. 2). Palaeontological control on the age of the palaeomagnetically sampled interval is less precise than is available for the immediately underlying rocks (Fig. 2). However, the stratigraphic interval in which the chamositic ironstones are developed is relatively thin (100 m) compared with the thickness of the conformably underlying well-dated Givetian-Frasnian part of the succession (approximately 600 m). This consideration, together with the presence immediately above the middle ironstone unit of the classically Middle to Upper Devonian fossil plant *Barinophyton* (Fig. 2), suggests that the palaeomagnetically sampled interval is probably wholly of Upper Devonian age.

The sampled ironstone units at Kurāgh are relatively thin and are intercalated with micritic and dolomicritic carbonates and quartzose sandstones and siltstones (Fig. 2). The ironstone units are typically of a dark red-brown or pale yellowish-green colour and are characteristically massive and very fine grained, save that oolites and pisolites occur in many places with varying abundance over the whole range from complete absence to tightly-packed concentrations. Preliminary thin section studies of these rocks together with reconnaissance electron microprobe and XRF analyses of their composition indicates that they are texturally and mineralogically similar to the "sedimentary ironstone formations" of other authors [51–54], and for which a primary sedimentary origin is well established. The gross facies context of the Kurāgh ironstones, as reflected in the lithology, sedimentary structures, and fossil content of the associated sediments (Fig. 2), suggests very shallow marine or strandline depositional environments, periodically interrupted by interludes of non-marine conditions and emergence, and is consistent with the inferred environmental setting of sedimentary ironstone formations as described elsewhere [53–56]. Chamosite is the predominant mineral present in the Kurāgh rocks, both in the oolites and pisolites and in the intervening matrix. No major secondary alteration effects are evident in the textures or mineralogy of the rocks and their suitability for palaeomagnetic work therefore follows.

The samples were oriented with a Brunton compass and then extracted as blocks of large (decimetre-scale) size. Bedding of the ironstone units was in most cases best determined from the immediately enclosing sedimentary rocks, the average attitude of which throughout the sampled interval is uniformly towards the northwest at about 50°.

4. Laboratory treatment

Most of the samples are of a pisolitic nature (Table 1). In order to minimize the possible disturbing effects of inhomogeneous magnetization upon the measured direction of remanence [57], we studied several specimens from each field sample. Altogether 118 specimens, each of 2.5 cm diameter and 2.25 cm length, were obtained in the laboratory from the field samples using a drill-press. From each field sample

TABLE 1

Analytical data from laboratory processing of the samples

Stratigraphic location (cf. Fig. 2)	Sample No.	Bulk colour and petrographic character	Natural remanent magnetization results				After thermal or AF demagnetization (mean directions)						
			number of specimens	intensity [a]	induced [b]	Q-value	number of specimens	before bedding correction		after bedding correction		k	α_{95} (°)
								D(°)	I(°)	D(°)	I(°)		
Unit 3	C106	mid-green pisolitic	23	0.17–12.60	3.08–11.09	0.05–1.14	22	317.5	+39.5	317	−2	20	7
	C104	red, non-pisolitic	18	13.54–21.45	3.63–6.11	2.64–5.15	16	331.5	+35	330	−17.5	34.5	6.5
	C103	red, non-pisolitic	4	14.93–27.70	4.17–4.44	3.58–6.23	4	296.5	+62	308	+10	24	19
Unit 2	C92	dark green pisolitic	21	0.43–70.62	10.66–20.13	0.04–3.50	11	114	−58	129	+2	21.5	10
	C91	dark green pisolitic	15	0.43–9.86	7.17–16.08	0.08–0.61	10	92	−41.5	106.5	+7.0	24	9.5
Unit 1	C77	dark green pisolitic	27	0.16–5.54	10.23–13.96	0.01–0.54	27	349	+56.5	345	−3	112.5	2.5
	C75	red and green pisolitic	11	0.66–5.67	4.72–6.16	0.12–0.98	10	332.5	+40	332	−21	16	12.5

Mean sample directions							Pole positions				
cleaning	number of specimens	D(°)	I(°)	k	α_{95} (°)	location	longitude	latitude	dp	dm	palaeolatitude
All therm. [c]	7	326	−9.5	15.5	16	72.5°E, 36.2°N	117.5°E	38°S	8°	16°	6.5°S
All AF [d]	7	317.5	−5	14.5	16.5	as above	127.5°E	34.5°S	8.5°	16.5°	2.5°S
All AF + therm.	7	318	−6.5	14	16.5	as above	126°E	34.5°S	8.5°	16.5°	3°S

[a] Initial intensity of natural remanent magnetization (in mA m^{-1}).
[b] Initial intensity of induced magnetization (in mA m^{-1}).
[c] Thermally demagnetized specimens only.
[d] Alternating-field demagnetized specimens only.

two to four pilot specimens (25 in total) were subjected to progressive thermal demagnetization treatment in 12 to 16 steps up to 685°C in a furnace described by McElhinny et al. [58]. Another 66 specimens were subjected to partial progressive thermal demagnetization in 8 to 10 steps over a selected temperature interval. Extreme care was taken to keep the specimens in a very much reduced magnetic environment between each phase of thermal treatment in order to avoid excessive buildup of a viscous remanent magnetization (VRM). All heatings were done in air.

In addition, 27 specimens were subjected to progressive alternating-field (AF) demagnetization in 9 to 17 steps up to 210 mT in a two-axis tumbler modified after Evans [59].

All magnetic measurements were taken on a ScT superconducting rock magnetometer which had been interfaced with a Digico minicomputer for immediate data reduction.

5. Results and interpretation

The initial measurements showed for most samples rather well concentrated directions in the northwestern quadrant with low downwards dip (Fig. 3A). The initial remanence directions are not far from, but certainly do not coincide with, the present local field direction in the study area ($D = 2.7°, I = +52°$). Only the two more coarsely pisolitic samples (i.e., samples C91 and C92, both from the lowermost of the three ironstone stratigraphic units studied) show a streaking of directions towards the southeastern quadrant and upward-pointing (Fig. 3A), being suggestive for reversed directions (Fig. 3C).

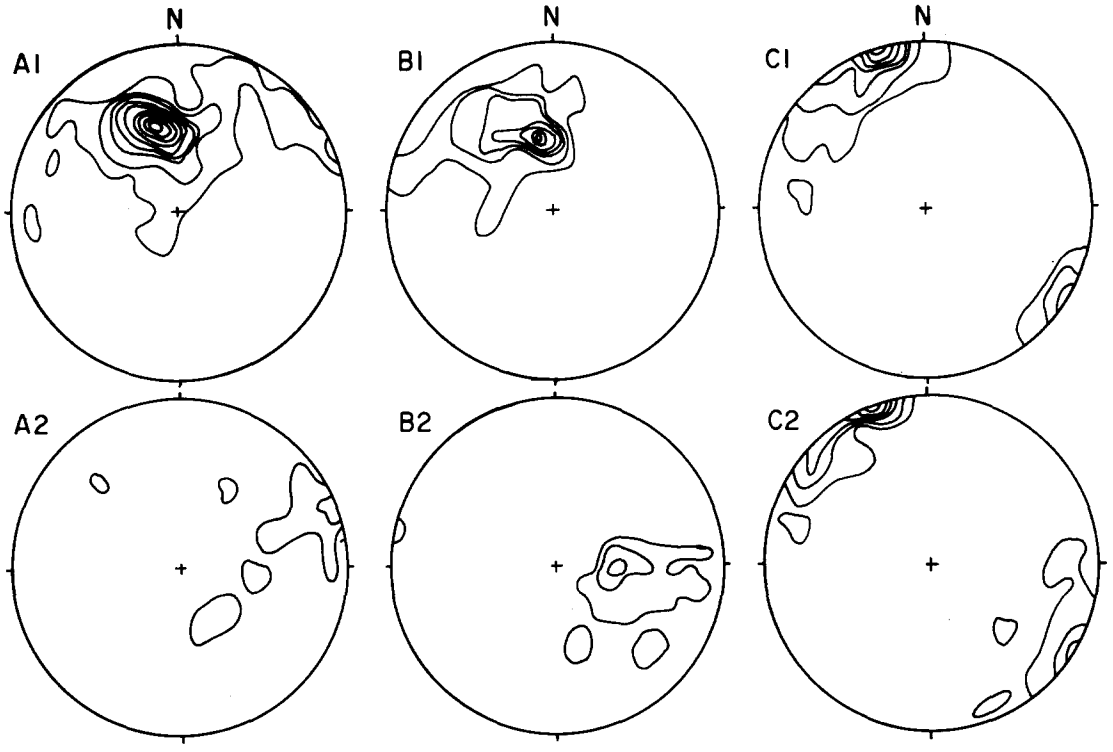


Fig. 3. Frequency distribution of natural remanent magnetization directions in equal-area projection. Subscript "1" denotes lower hemisphere projection, subscript "2" denotes upper hemisphere projection. Frequency interval = 3. A. Initial specimen directions. B. Characteristic magnetization directions after thermal or alternating-field cleaning, shown with respect to the present horizontal. C. Characteristic magnetization directions after thermal or alternating-field cleaning, dip correction applied.

The induced magnetization (0.1 mT applied field) was in general up to one order of magnitude higher in intensity than the initial remanent magnetization, except for the red ironstone samples C103 and C104 (Table 1) which also had a much lower susceptibility.

Two magnetic components could be distinguished upon thermal demagnetization. A recent local field component which comprised as much as 75% of the initial intensity (specifically in samples C92, C103 and C104; see for example Figs. 4B and 5A) became eliminated completely between 200° and 300°C

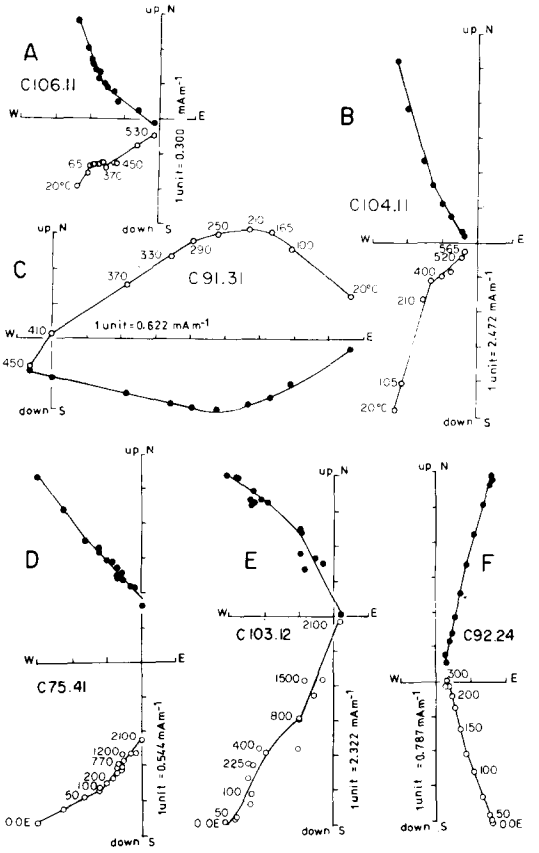


Fig. 4. Demagnetization diagrams of representative specimens treated by thermal methods or with alternating fields. The points denote successive positions (in orthogonal projection) of the end of the resultant magnetization vector during progressive demagnetization. Dots denote projections on the vertical east-west plane. Open circles denote projections on the horizontal plane. Numbers denote successive peak values of the applied temperatures (diagrams A, B, C) or millitesla (mT) peak values of the applied alternating fields (diagrams D, E, F). Sample identification: C106.11 = field sample C106, laboratory specimen 11.

except in the case of the two red ironstone samples C103 and C104 where elimination took place between 300° and 400°C. Above these temperatures only the characteristic magnetization component

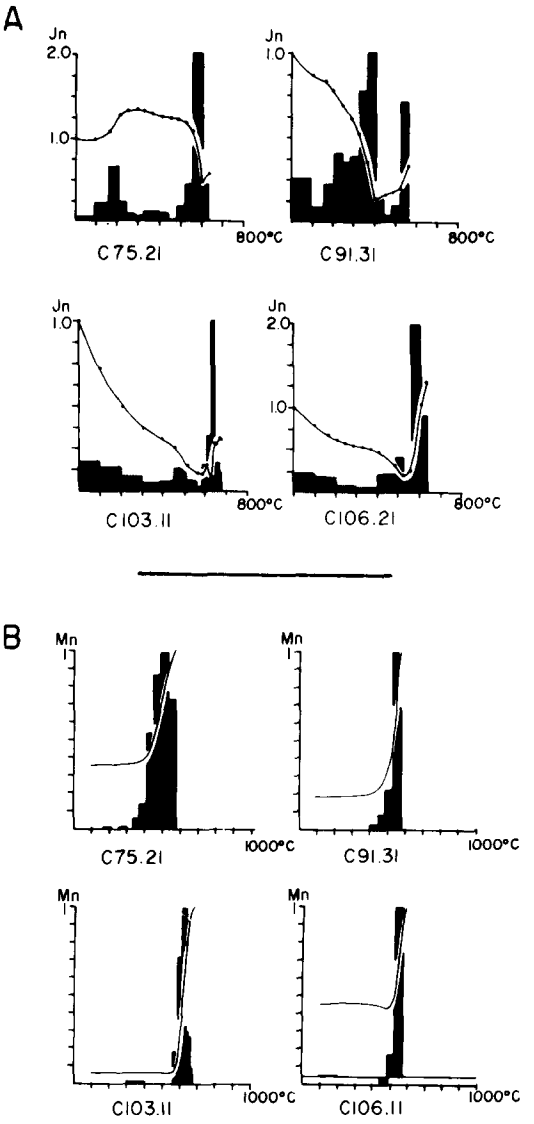


Fig. 5. A. Normalized intensity decay curves of remanent magnetization during thermal treatment of some representative specimens; histograms show blocking temperature spectra. Note the increase in remanence because of a VRM build-up at the higher temperatures. B. Normalized intensity curves and histograms of the induced magnetization of some representative specimens indicating the increase in susceptibility after treatment at higher temperatures. Sample identification code as for Fig. 4.

became eliminated and its direction could be determined accurately. For most of the samples this occurred between 300° and 500°C, and for the red ironstone samples between 300° and 600°C (Figs. 4A–C and 5A).

The specimens became gradually more prone to VRM buildup as can be inferred from the large increase in susceptibility above 400°C (respectively 550°C for samples C103 and C104; Fig. 5B). This VRM buildup became excessive after elimination of the characteristic magnetization component and prevented further analysis after heating to temperatures above 500–600°C.

Most specimens showed a tendency for the magnetization vector-path to bypass the origin (Figs. 4A–C). This effect can be attributed largely to a distinct VRM component, and also to inhomogeneous magnetization of the pisolitic samples [57]. For such specimens, the direction of the characteristic magnetization component could be determined accurately from the orthogonal projection figures [60, 61]. Only a few of the specimens had to be rejected because of uninterpretable behaviour.

Directional results obtained after AF demagnetization and after thermal treatment are in good agreement (Table 1). A recent local field component, where recognizable, was eliminated at about 20 mT peak value. The characteristic magnetization component became eliminated separately at the higher AF-demagnetization steps (Figs. 4D–F). The good directional agreement with the thermally obtained results indicates an absence of VRM effects in the determination of the latter (i.e., the thermally obtained results). The orthogonal projection figures obtained from the AF treatment showed the same bypassing effect as in the case of the thermally obtained results, indicating the effect of inhomogeneous magnetization (Figs. 4D–F).

Both normally magnetized (northwesterly and downward-pointing for samples C75, C77, C103, C104, C106) and reversely magnetized directions (southeasterly and upwards pointing for samples C91 and C92) are present (Figs. 3B and 6A). After dip correction all directions concentrate in the horizontal plane (Figs. 3C and 6B). As similar dip corrections were applied to all samples, there is no improvement of grouping.

Because of the primary sediment nature of the pisolitic ironstones, the age of the characteristic magnetization can be taken as identical with the chronostratigraphic age of the rocks.

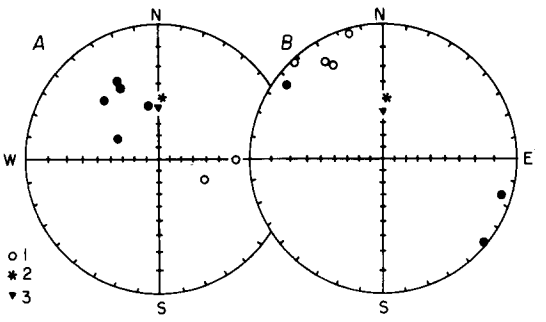


Fig. 6. Equal-area projection of sample-mean directions after thermal or alternating-field treatment, shown with respect to the present horizontal (A), and after dip correction (B). 1 = sample-mean direction; 2 = present local field direction at the sampled area; 3 = expected present local field direction according to the axial geocentric dipole field formulae. Open symbols denote upward-pointing directions. Closed symbols denote downward-pointing directions.

6. Comparison with Gondwanaland and with Asia

Most reconstructions of Pangea's configuration during the mid-Palaeozoic [62–65] figure a wide eastward-opening Tethys, with the Indian part of Gondwanaland situated at moderate southern latitudes and with Asia situated at moderate northern latitudes. The mid-Palaeozoic palaeomagnetic data from both continents would therefore be expected to be quite distinct in inclination. This circumstance is clearly favourable for resolving the affinity of the present palaeomagnetic data from the Upper Devonian rocks of Chitral with either the Indian or the Asian group of data.

Although Devonian palaeomagnetic data are not yet available from the Indian subcontinent [5,10], India's Late Devonian latitudinal position and orientation can nevertheless be determined from the mean Gondwanaland palaeomagnetic data for the period involved with the continents reassembled in the Smith and Hallam [25] configuration. Recently noted imperfections of this reconstruction, such as a possible wider spacing between the various continental fragments [66], the former position of parts of Iran and Afghanistan, the disputed relative position of East and West Gondwanaland [5,67], and the evi-

dent more westward palaeoposition of Australia relative to Antarctica [26,68,69], can be regarded as second-order effects in terms of the present exercise and can therefore be neglected here. Palaeomagnetic data from Upper Devonian rocks of Gondwanaland are few, and most of those available come from the Australian continent. Accordingly, it is expedient in the present analysis to combine all Middle Devonian to Early Carboniferous Gondwanaland data together (Table 2), even though these data do not form a particularly tight grouping on the Smith and Hallam reconstruction probably because Gondwanaland may have undergone some movement during this time interval.

Many palaeomagnetic data are available from the Upper Devonian of the Siberian platform, but for reasons of compatibility with the Gondwanaland data, we have also combined available data from Middle Devonian to Early Carboniferous rocks of this region (Table 2). There is a noticeable difference between mean results according to data from Khramov's [70] various lists (compiled by Irving et al. [71] and McElhinny et al. [72]). As Khramov's 1975 list may indicate an improvement in results compared with his 1971 list, means from both these lists are given separately (Table 2).

The present results indicate a Late Devonian palaeolatitude for the Chitral area at about 3° south (Fig. 7). If the Chitral area were originally either a part of the Indian fragment of Gondwanaland or a separate Gondwana microcontinent, in each case

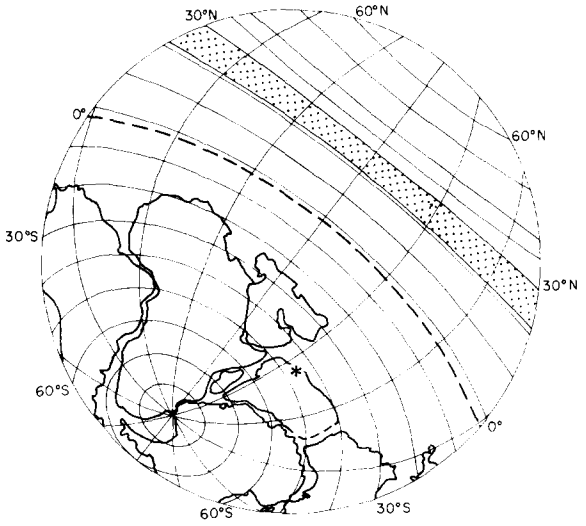


Fig. 7. Observed and expected latitudinal positions of the sampled area in linear polar projection. Dashed line: palaeolatitudinal position at 3°S. Asterisk: expected latitudinal position if the sampled area formed part of a subcontinental Gondwanic fragment similar to the proposed Greater India of Veevers et al. [20]. Gondwanaland (cf. Smith and Hallam [25]) repositioned according to the mean Middle Devonian to Early Carboniferous pole. Stippled area: expected palaeolatitudinal position of the sampled area assuming it has formed part of Asia since the Devonian without displacement relative to the Siberian platform.

with its present position relative to India being approximately maintained, then a palaeolatitude of about 45° south is to be expected in the Smith and Hallam reassembly (Fig. 7). This palaeoposition con-

TABLE 2

Mean south pole positions for the Middle Devonian to Lower Carboniferous

	N [a]	Longitude	Latitude	k	α_{95} (°)	Expected direction [b]	
						D(°)	I(°)
Gondwanaland [c]	8	35.9°E	25.6°S	14.5	15	34.7	−34.9
Chitral	–	126°E	34.5°S	–	–	(318	−6.5)
Siberian platform [d]	32	43.8°W	26.5°S	7.5	10	81.1	+55.2
Siberian platform [e]	9	48.4°W	7.1°N	6.6	25.3	115.2	+35.7

[a] Number of palaeomagnetic determinations.
[b] Directions expected in Chitral according to the cited pole, assuming that Chitral formed part of the respective continents indicated without subsequent tectonic disturbance.
[c] South pole with respect to Africa.
[d] According to Khramov [70, Issues 1 and 2] as quoted by Irving et al. [71] and McElhinny et al. [72].
[e] According to Khramov [70, Issue 3] as quoted by Irving et al. [71] and McElhinny et al. [72].

trasts strongly with the 20° to 30° northern palaeolatitude which is to be expected for the Chitral area in the period Middle Devonian–Early Carboniferous assuming no relative latitudinal displacement of the area with respect to the Siberian platform (Fig. 7).

7. Discussion and conclusions

As indicated in Fig. 7, the Late Devonian position of the Chitral area conforms better with the contemporary palaeoposition of Asia (as determined from the Siberian platform data) than with that of Gondwanaland. This relationship is further enhanced if one straightens out the Hindukush-Pamir-Karakoram syntaxial bend (cf. Carey [73] and Powell [23]), the northward-convex shape of which is evidently related to collision tectonics, and assumes the Chitral area to have been shunted northwards by the Indian continental block during the first phase of its post-collision convergence with Eurasia beginning probably about 50 m.y. B.P. [11,18,23,74,75]. ERTS photographs and model studies [74,76–78] indicate that the Indian continental block may have indented southern Asia, possibly by as much as 1500 km, through sidewise movement of crustal blocks throughout an extensive region of central Asia north of the collision zone. Molnar and Chen [75] have recently interpreted the preliminary palaeomagnetic data from south-central Tibet [45] as additional support of this suggestion, and cite a palaeomagnetically deduced crustal shortening in excess of 2500 km. Powell's [23] suggestion that the Chitral area formerly occupied a position further to the south and aligned with the Makran-Zagros zone (cf. [73]) leads to a northward displacement of similar magnitude. Such a southern palaeolatitude of the Chitral area would effectively remove the apparent discrepancy between the palaeolatitude of the Chitral area computed on the basis of the present samples and its expected palaeolatitude assuming the area to have been part of Asia since at least the Devonian with no subsequent displacement relative to the Siberian platform (Table 2).

This palaeomagnetic argument is further substantiated by palaeontological data [33–35] which demonstrate distinct provincial differences between marine benthonic faunas of contemporary age in Chitral and in areas of the Indian subcontinent south of the Axial Ophiolite Belt. The combined palaeomagnetic and palaeontological evidence therefore indicates that the Chitral area is an original part of Asia and does not have Gondwanic affinities. This conclusion does not support Stöcklin's [12] and Andrieux and Brunel's [43] speculated eastward continuation through the eastern Hindukush and Karakoram regions of the belt of Gondwanic microcontinents that are known to occur in Iran and Afghanistan. The speculated development of the zone still further east in southern and/or northern Tibet can not presently be resolved on the basis of the existing palaeomagnetic data from that area. To date, the only palaeomagnetic data that are available from the latter region, i.e. from Ladakh [79] and from south-central Tibet [45] are not capable of deciding the issue.

The aberrant declination of the present results from Chitral ($D = 318°$) indicates a counter-clockwise rotation of the study area of about 40° with respect to present-day north (and in excess of 100° with respect to the Devonian palaeonorth direction of the Siberian platform; see Table 2). This counterclockwise rotation, which is consistent with Powell's [23] model of the collision tectonics of the region, has evidently resulted from the major thrusting that is conspicuous in the area and which clearly is associated on a more regional scale with the formation of the Hindukush-Pamir-Karakoram syntaxial bend. A counter-clockwise rotation ($D = 311°$) was also found by Burtman and Gurariy [80] in Lower Cretaceous rocks from the exterior zone of the western Pamirs (i.e., from the western flank of the Northern Pamir of Fig. 1A), adding further to the interpretation that these appreciable counter-clockwise rotations reflect the formation of the western limb of this regional syntaxis. The apparently aberrant declinations of the Upper Palaeozoic to Lower Mesozoic Panjal Traps in Kashmir [81] may be interpreted as a contrasting clockwise rotation of the Panjal Nappe in this area [5] since this nappe forms part of the other (eastern) limb of this conspicuous syntaxial feature. Recent palaeomagnetic results from the Ladakh batholith further north in the Axial Ophiolite Belt (near Kargil) also indicate a clockwise rotation of the area [79] and can be similarly interpreted.

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